

FROM ZERO TO HERO: COMPLETE LEARNING ROADMAP

Mastering Node.js, PostgreSQL, Vanilla Web & Kotlin to Build Full-Stack Systems

CURRICULUM DESIGN: 8-Week Progressive Learning Path for Final Year Projects & Career Readiness

1. Why This Specific Tech Stack Matters

Building an enterprise project without relying on bulky all-in-one frameworks (like React, Angular, or Django) forces you to master core computer science and software engineering fundamentals: protocols, DOM architecture, relational calculus, asynchronous event loops, and native mobile rendering. This knowledge transfers directly to any future language or framework.

1. RELATIONAL DATA

PostgreSQL, 3NF, Keys, ACID transactions

2. REST ARCHITECTURE

Node.js, Express, JWT, RBAC security

3. FRAMEWORK-FREE WEB

HTML5, CSS3 Grid, Vanilla JS SPAs

4. NATIVE MOBILE

Kotlin, Retrofit2, Material 3, Android SDK

2. Phase 1: Foundations & Relational Database Mastery

STAGE 1: RELATIONAL DATABASE MODELING & POSTGRESQL (SQL)

[Timeline: Week 1 (Days 1 - 7)]

Core Topics to Master:

Relational theory, 1NF/2NF/3NF normalization, Primary/Foreign Keys, ON DELETE CASCADE vs RESTRICT, indexes, joins (INNER, LEFT), subqueries, and ACID transaction rules.

Hands-on Coding Milestone:

Design `schema.sql`: Create `users`, `books`, and `borrow_records` tables with constraints and foreign key linkages.

STAGE 2: JAVASCRIPT FUNDAMENTALS & ASYNCHRONOUS PROGRAMMING (ES6+)

[Timeline: Week 2 (Days 8 - 14)]

Core Topics to Master:

Arrow functions, destructuring, CommonJS modules, Promises, `async/await`, Array methods (`map`, `filter`, `reduce`), error handling with `try/catch`, and the Node.js Event Loop.

Hands-on Coding Milestone:

Write a standalone Node script that connects to PostgreSQL and performs asynchronous CRUD operations.

3. Phase 2: Backend RESTful API & Security Engineering

STAGE 3: NODE.JS & EXPRESS RESTFUL API DEVELOPMENT

[Timeline: Week 3 (Days 15 - 21)]

Core Topics to Master:

HTTP methods (GET, POST, PUT, DELETE), status codes (200, 201, 400, 401, 403, 404, 500), Express Router modularization, body parsing (`express.json`), and error middleware.

Hands-on Coding Milestone:

Build `/api/books` and `/api/borrow` endpoints with input validation and clean JSON responses.

STAGE 4: AUTHENTICATION, JWT TOKENS & ROLE-BASED ACCESS CONTROL

[Timeline: Week 4 (Days 22 - 28)]

Core Topics to Master:

Stateless session management, password hashing with `bcryptjs` (salt rounds), signing & verifying JSON Web Tokens (`jsonwebtoken`), Bearer headers, and role guards.

Hands-on Coding Milestone:

Implement secure login, student registration, and guard circulation routes against unauthorized student requests.

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Section II: Advanced Backend, Vanilla Web & Native Android Mastery

PHASES 3 & 4: External Integrations, Framework-Free SPA & Mobile App Engineering

4. Phase 3: Advanced Integrations (Excel & External APIs)

STAGE 5: EXTERNAL API INTEGRATION (OPEN LIBRARY BY INTERNET ARCHIVE)

[Timeline: Week 5 (Days 29 - 35)]

Core Topics to Master:

Third-party HTTP requests using Native Fetch / Axios, User-Agent policies, rate limits, parsing nested external JSON structures, image URL construction, and ISBN normalization.

Hands-on Coding Milestone:

Build `openLibraryService.js` to proxy live searches to `openlibrary.org` and enable 1-click book importing.

STAGE 6: BULK SPREADSHEET PROCESSING WITH SHEETJS (XLSX) & MULTER

[Timeline: Week 6 (Days 36 - 42)]

Core Topics to Master:

Multipart/form-data file uploads with Multer, memory buffer manipulation, parsing Excel (.xlsx, .csv) workbooks, flexible column mapping, and duplicate prevention in batches.

Hands-on Coding Milestone:

Implement `/api/auth/students/upload` for batch roster uploads and `/api/auth/students/template` for binary downloads.

5. Phase 4: Client Development (Vanilla Web & Native Android)

STAGE 7: LIBRARIAN ADMIN WEB PORTAL (VANILLA HTML5, CSS3 & JAVASCRIPT)

[Timeline: Week 7 (Days 43 - 49)]

Core Topics to Master:

Semantic HTML5, CSS Grid & Flexbox, CSS Variables, DOM manipulation (`document.querySelector`, `addEventListener`), dynamic tables, custom modals, Drag-and-Drop file API, and SPAs.

Hands-on Coding Milestone:

Construct the complete Admin Portal with Dashboard tabs, Live Search modals, Circulation Desk, and Student roster.

STAGE 8: STUDENT MOBILE APPLICATION (NATIVE ANDROID WITH KOTLIN)

[Timeline: Week 8 (Days 50 - 56)]

Core Topics to Master:

Kotlin syntax (data classes, null safety, coroutines), Android Activity lifecycle, ViewBinding, Material 3 Design, Retrofit2 & OkHttp3 networking, and SharedPreferences session caching.

Hands-on Coding Milestone:

Build the Android Student App with Login, Catalog search with Category Chips, Shelf tags, and My Loans tracker.

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Section III: Recommended Free Resources, Practice Projects & Checklist

SELF-ASSESSMENT: Step-by-Step Milestones to Build Your Own Enterprise Project

6. Curated High-Yield Free Learning Resources

You do not need paid courses to master this stack. Leverage these top-tier official documentations and tutorials:

PostgreSQL & SQL	PostgreSQL Official Docs (postgresqltutorial.com)	Interactive SQL exercises on queries, schema design, and indexes.
Modern JavaScript (ES6+)	javascript.info & MDN Web Docs	The gold standard for understanding Promises, async/await, and the DOM.
Node.js & Express	nodejs.org/docs & expressjs.com guide	Step-by-step guides for routing, middleware, and request/response lifecycle.
JWT & Web Security	jwt.io/introduction & OWASP API Top 10	Best practices on token signing, password hashing, and endpoint protection.
SheetJS & File Processing	docs.sheetjs.com	Comprehensive guide to reading, writing, and manipulating Excel workbooks in JS.
Open Library API	openlibrary.org/developers/api	Documentation for search.json, isbn.json, and the Covers API.
Kotlin & Android Basics	developer.android.com/courses	Google's official Android Basics with Kotlin course (free with code labs).
Retrofit2 Networking	square.github.io/retrofit	Official documentation for type-safe HTTP client communication in Android.

7. Progressive Mini-Projects (Build Up to the Final Project)

Do not attempt to build the entire system all at once. Build these 4 progressive stepping-stone projects:

1. CLI Book Tracker (Node.js + PostgreSQL)

Create a terminal app that adds books, lists them from PostgreSQL, and deletes them. Teaches SQL queries and Node.js db connection.

2. RESTful Auth Service (Express + JWT + bcrypt)

Build an API with register, login, and protected /profile route. Teaches passwords hashing, token generation, and middleware guards.

3. Web Circulation Dashboard (Vanilla HTML/CSS/JS)

Build a single-page web app with fetch() to list books, search dynamically, and submit forms without refreshing the browser.

4. Android Catalog Viewer (Kotlin + Retrofit)

Build a mobile app with RecyclerView to fetch and display the book catalog from your local Node.js API with cover images.

8. Self-Readiness Checklist Before Building

- ☐ I understand how foreign keys, cascades, and indexes work in PostgreSQL.
- ☐ I can write async/await functions and handle rejected promises gracefully in Node.js.
- ☐ I can explain what a JWT header, payload, and signature are and why they are stateless.
- ☐ I know how to manipulate the DOM with document.createElement and fetch() without React.
- ☐ I understand Android Activities, Retrofit ApiService interfaces, and Kotlin coroutines.